Report Date:January 28, 2019 File:120

Report Number:115167

Town of Ladysmith

PO Box 220, 410 Esplanade
Ladysmith BC V9G 1A2

Dear Town of Ladysmith

Re: Non-compliance Advisory Letter, Permit 120, Ladysmith, BC, Effluent

On January 28, 2019, Ministry of Environment and Climate Change Strategy, Environmental Protection Division staff conducted an inspection of your facility, Town of Ladysmith wastewater treatment plant located at Ladysmith, BC with authorization number 120 under the *Environmental Management Act*.

This Advisory, the alleged violations and the circumstances to which it refers will form part of the compliance history of Town of Ladysmith, and will be taken into account in the event of future non-compliance.

Please note that this authorization is considered to be out of compliance until such a time as it can be confirmed to meet the authorization requirements.

Inspection Details:

Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.1: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.1 The maximum rate at which effluent may be discharged through the sewage treatment plant is 6 100 m3/d; the annual average rate of discharge is 3 050 m3/d. Flow in excess of treatment plant capacity is to be discharged to the bypass outfall. The maximum rate at which effluent may be bypassed is 8 500 m3/d; the total annual volume which may be discharged to the bypass outfall is 228 000 m3.

Ministry of Environment and Climate Change Strategy

Compliance
Environmental Protection Division

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Details/Findings:	On January 28, 2019, Ministry of Environment and Climate Change Strategy (Ministry) Officer Darren Stewart (Officer Stewart) conducted an office review inspection of the Town of Ladysmith wastewater treatment facility located in Ladysmith, BC to verify compliance with Permit PE-120. Permit 120 authorizes the discharge of effluent to Ladysmith Harbour from a municipal sewage system. The Permit was first issued on March 12, 1965, and last amended on January 10, 2994. The inspection period for this report is from January 01, 2018 to December 31, 2018 and included a review of the following documents: - 2018 Quarterly Water Quality Data Reports (2018 Q1, Q2, Q3, Q4 WQ DATA) - 2018 outfall inspection report submitted to the Ministry on September 13, 2018 - Trigger Event Report for December 20, 2018 event, submitted to the Ministry on December 27, 2018 - 2018 yearly bypass report submitted to the Ministry on January 14, 2019 The following 13 exceedances of the maximum rate at which effluent may be discharged through the sewage treatment plant were noted on the following dates as reported in the quarterly WQ DATA: January 18, 2018 = 6904 m3 January 21, 2018 = 7966 m3 January 22, 2018 = 6179 m3 January 27, 2018 = 6951 m3 January 28, 2018 = 9093 m3 January 29, 2018 = 11475 m3 January 30, 2018 = 6191 m3 December 13, 2018 = 6361 m3 December 16, 2018 = 6117 m3 December 18, 2018 = 6918 m3 December 20, 2018 = 7747 m3 December 22, 2018 = 6617 m3 December 23, 2018 = 7444 m3 There was one daily report of flows discharged to the bypass outfall, which occurred on December 20, 2018 (721 m3). The compliant 2018 annual average rate of discharge was calculated by Officer Stewart as 2824 m3/d.
Compliance:	Out
Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.2: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.2 The characteristics of the discharge shall not exceed: Sewage Treatment Plant Discharge: 5-day Biochemical Oxygen Demand: 130 mg/L. Total Suspended Solids: 130 mg/L. Overflow Bypass Discharge: Screened combined stormwater and sewage effluent.

Details/Findings:	There were no exceedances reported for characteristics of the discharge during the inspection period covered by this report. All reported values during the inspection period for 5-day Biochemical Oxygen Demand (BOD5) and Total Suspended Solids (TSS) were well below permitted limits, The maximum values reported during 2018 were BOD5 = 20 mg/L (May) and TSS = 22 mg/L (August).
Compliance:	In
Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.3: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.3 The works authorized are influent screening, a Spiragester primary type treatment plant, chlorination and dechlorination facilities, autothermophilic aerobic sludge processing facilities, foul air scrubber, sludge dewatering facilities, overflow bypass, bypass screening and solids removal facilities, 300 mm and 710 mm diameter parallel outfalls extending 850 m from mean low water to a depth of 17.9 m below mean low water, and related appurtenances approximately located as shown on the attached Site Plan A.
Details/Findings:	The authorized works were not evaluated as part of this data review inspection; therefore, compliance with this requirement was not applicable.
Compliance:	Not Applicable
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.1 Maintenance of Works and Emergency Procedures** 2.1: The Permittee shall inspect the pollution control works regularly and maintain them in good working order. In the event of an emergency or condition beyond the control of the Permittee which prevents continuing operation of the approved method of pollution control, the Permittee shall immediately notify the Regional Waste Manager and take appropriate remedial action.

Details/Findings:	On December 20, 2018, as reported by Town of Ladysmith, there was an emergency condition (power surges and outage) caused by a condition beyond the control of the Permittee (wind storm) that prevented continuing operation of the approved method of pollution control. The Permittee immediately notified Emergency Management BC by calling the Spill Reporting Hotline (DGIR# 183468) and took appropriate remedial action as outlined in the Trigger Event Report provided to the Ministry on December 27, 2018. The report described a bypass of 153 m3 of screened sewage (heavily diluted by rain water) through our overflow bypass, as well as 568 m3 that did not receive secondary treatment - flowing through screening and the spirogester without receipt of disinfection. An on site electrician, remote assistance from an electrical engineer and on site technicians stopped the bypass after approximately 2.5 hours.
Compliance:	In
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.2 Bypasses** 2.2: The discharge of effluent which has bypassed the designated treatment works is prohibited unless the consent of the Regional Waste Manager is obtained and confirmed in writing.
Details/Findings:	On February 21, 2018 (DGIR#174028) the town of Ladysmith completed a dye-test inspection of their wastewater treatment plant outfall during which dye was observed bubbling out of the ground during low tide, approximately 110 m from the shore. On February 22 it was confirmed that the leak was occurring from the 300mm AC (asbestos-cement) original outfall line. The volume and duration of this bypass of the designated treatment works was reported as unknown. The last outfall inspection prior to the one described in this section that observed the leak occurred in 2011.
Compliance:	Out
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.5 Outfall Inspection** 2.5: The Permittee shall conduct a dye test on the outfall lines (or inspect by another method approved by the Regional Waste Manager) every five years or as may otherwise be required by the Regional Waste Manager.

Details/Findings:	The "Ladysmith Wastewater Treatment Plant Outfall – 2018 Inspection Report" written by Great Pacific Engineering and Environment, was submitted to the Ministry by the Town of Ladysmith on September 13, 2018. The report confirms that both outfalls were inspected on February 21, 2018 using a dye test.
Compliance:	In
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.6 Chlorination** 2.6: Maintain a chlorine residual (at the point of discharge or prior to dechlorination) between 0.1 and 1.0 mg/L at all times.
Details/Findings:	Chlorine residual values submitted in quarterly WQ data to the Ministry confirmed that a chlorine residual (at the point of discharge or prior to dechlorination) was maintained between 0.1 and 1.0 mg/L at all times.
Compliance:	In
Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.1 Discharge Monitoring, 3.1.1 Sampling and Analysis** 3.1.1: The Permittee shall install a suitable sampling facility and obtain a composite sample of the effluent from the treatment plant effluent once every month. The sample is to consist of four grab samples taken over a two-hour period at maximum flow and mixed to form a single sample for subsequent analysis. Proper care should be taken in sampling, storing and transporting the sample to adequately control temperature and avoid contamination, breakage, etc. Obtain analyses of the sample for the following: 5-day Biochemical Oxygen Demand; Total Suspended Solids.
Details/Findings:	This data review inspection confirmed monthly analyses was performed of samples for 5-day Biochemical Oxygen Demand and Total Suspended Solids as required, in addition to fecal coliforms. Sample analysis was performed by Maxxam Analytics of Burnaby, BC.
Compliance:	In

Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.1 Discharge Monitoring, 3.1.2 Flow Measurement** 3.1.2: Provide and maintain suitable flow measuring devices and record once per day the effluent volume discharged over a 24-hour period from the sewage treatment plant and the overflow bypass.
Details/Findings:	The effluent volume discharged over a 24-hour period from the sewage treatment plant and the overflow bypass are contained within the quarterly WQ DATA reports submitted to the Ministry.
Compliance:	In
Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.4 Reporting** 3.4: Maintain data of analyses and flow measurements for inspection and every three months submit the data, suitably tabulated, to the Regional Waste Manager for the previous quarter. The first report is to be submitted by January 31, 1994. Based on the results of the monitoring program, the Permittee monitoring requirements may be extended or altered by the Regional Waste Manager.
Details/Findings:	Officer Stewart confirmed all 2018 data of analyses and flow measurement are submitted to the Ministry as required by this section.
Compliance:	In

The scope of this inspection was a monitoring and reporting data review only, completed for reporting purposes related to the annual Conditional Management Plan (CMP) which is to provide enhanced management of the molluscan shellfish harvest in waters adjacent to the Town of Ladysmith wastewater treatment facility.

There were two trigger events reported during the inspection period:
February 21, 2018 (DGIR 174028) - an outfall inspection confirmed a leak in the outfall 110m from shore which was observed during low tide.
December 20, 2018 (DGIR 183468) - during a wind storm/power outage the discharge of screened sewage bypassed the works and discharge of effluent without receipt of secondary treatment or disinfection

Compliance History:
2018-01-23 - IR76276 - Advisory for discharge rate exceedances, characteristic (BOD5) exceedances, chlorine residuals outside of required range and a lapse of outfall inspection frequency.

Please submit all annual/quarterly/monthly reports and data submissions to the Ministry's Routine Environmental Reporting Submission Mailbox at:
EnvAuthorizationsReporting@gov.bc.ca

More information about the reporting requirements may be found at:
http://www2.gov.bc.ca/gov/content/environment/waste-management/waste-discharge-authorization/data-and-reportsubmissions/routine-environmental-reporting-submission-mailbox

More information about Environmental Compliance and how it is assessed can be found at:
https://www2.gov.bc.ca/gov/content/environment/natural-resource-stewardship/natural-resource-law-enforcement/environmental-compliance

Please be advised that this inspection report may be published on the provincial government website within 7 days.

If you have any questions about this letter, please contact the undersigned.

Yours truly,

Darren Stewart
Environmental Protection Officer

cc:

Attachments:

Deliver via:

Email: [X] **Fax:** [ ] **Mail:** [ ]
Registered Mail: [ ] **Hand Delivery:** [ ]

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Please note that sections of the permit, regulation or code of practice referenced in this inspection record are for guidance and are not the official version. Please refer to the original permit, regulation or code of practice.

To see the most up to date version of the regulations and codes of practices please visit
http://www.bclaws.ca

If you require a copy of the original permit, please contact the inspector noted on this inspection record.

It is also important to note that this inspection record does not necessarily reflect each requirement or condition of the authorization therefore compliance is noted only for the requirements or conditions listed in the inspection record.